

Troubleshooting

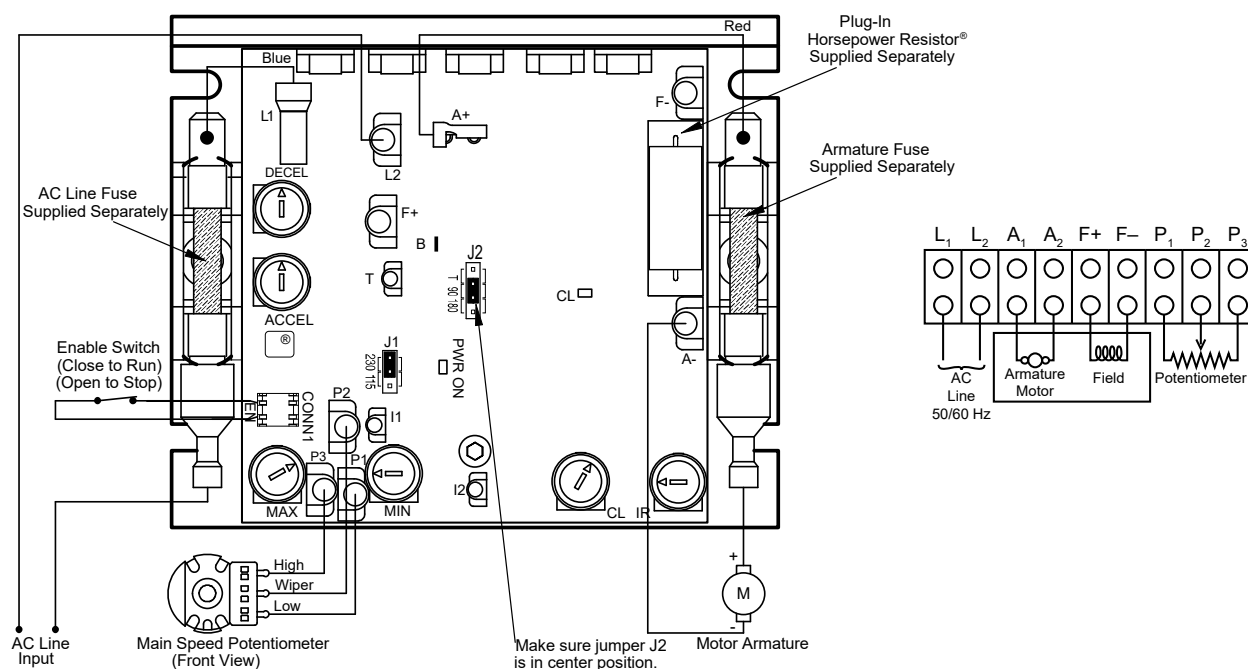
The following illustration shows the D.C. board housed in the conveyor Control Box. Some of the solutions to problems identified in the Troubleshooting table that follows refer to adjustments made by turning potentiometers on this board.

Basic KBMM™ Controller Board Connection Diagram

KBMM™ with Barrier Terminal Kit

CONTROL LAYOUT & GENERAL CONNECTION DIAGRAM (Model KBMM-225D Shown)

(Note: Control is set for 208 /230 VAC line input, 0-180 VDC output with armature feedback)



For more information refer to the KBMM™ *Installation and Operation Manual* (provided by the D.C. board manufacturer).

Problem	Possible Cause	Corrective Action
1. Motor does not run; Power On, Indicator not lit.	1. Power switch is in Off position, or 2. AC voltage is not brought to L1 and L2 terminals. 3. Blown line fuse	1. Toggle power switch to On position; or 2. Correct wiring to control box. 3. Replace line fuse with 20A rated 3AB-type fuse. If fuse blew due to mis-wiring, speed control module may be damaged.

Problem	Possible Cause	Corrective Action
2. Motor does not run; power On indicator lit.	1. Speed control knob set to 0 (zero). 2. Defective motor. 3. Plug-in Horsepower Resistor® not installed. 4. Blown armature fuse.	1. Turn knob clockwise to start motor. 2. Check for defective motor, worn brushes, etc. Replace motor 3. Install proper plug-in Horsepower Resistor®. 4. Replace fuse with fuse of proper value.
3. Motor hums or runs at very low speed (with speed control knob turned up to a high speed setting) or motor slows down substantially when load is applied.	1. Low voltage. 2. Overload condition: control in current limit mode (CL trim pot not set correctly). (CL LED is lit.) 3. Plug-in Horsepower Resistor® not correct size. 4. Incorrect wiring. Armature and shunt connections interchanged (shunt motor only).	1. Check line voltage at control and rewire as required. 2. Reduce loading; CL trim pot setting may need to be increased. See Adjustments section. 3. Install proper size plug-in Horsepower Resistor®. 4. Correct wiring (armature has lower resistance than field).

Problem	Possible Cause	Corrective Action
4. Erratic motor performance.	1. Worn or damaged motor: worn brushes, etc. 2. Overload condition. 3. Plug-in Horsepower Resistor® wrong size . 4. IR comp and/or CL trimpots not set properly. 5. Defective or damaged speed control module. 6. Dual Voltage Switch set in wrong position	1. Repair motor. 2. Remove overload. 3. Replace Plug-in Horsepower Resistor® with proper size. 4. Readjust trimpots. 5. Replace module. 6. Recheck line voltage and set Dual Voltage Switch to proper position, 115 or 230.
5. Motor continues to run when speed control knob is set to	1. Minimum speed trim pot not set to full counter-clockwise position. 2. IR comp trim pot set too high.	1. Readjust minimum speed trim pot. 2. Lower IR comp trim pot setting.
6. Motor runs in wrong direction.	Armature leads reversed.	Correct and reconnect armature leads.